

WATERWATCH · DESIGN DECISIONS REGISTER

WaterWatch design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/waterwatch		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-02	Amish Chadha	Register opened with the build plan; budget treated as a value-engineering target

WaterWatch design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md` , WWT-BLD-001) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Review the design-for-construction changes P1 to P13 (back plates on U-bolts, entries underneath, shield fixing, three optics holders, baffle on the back wall, lid layout, valve on the cell plate, pole-top mount and others)	Accept; or ask for changes	Accept	Every component and step of the build plan	WWT-DDR-003, Table 1
2	Shield fixing at public sites	(a) knurled thumb screws; (b) tamper-resistant M5 screws	(a) for the prototype; (b) for field units after the first site visit	Sun shield, step 18	WWT-DDR-003, A1
3	Antenna position	(a) whip pointing down from the bottom face; (b) short cable to an antenna on the pole top	(a), and measure signal strength at TRL 4	Enclosure bottom face, antenna	WWT-DDR-003, A2
4	Water volume used in the calculation	(a) keep the full-cavity 0.185 L, conservative for flushing; (b) revise to the standing volume of about 0.14 L now	(a), measure at TRL 4	None in the build; WWT-CAL-001 section D	WWT-DDR-003, A3
5	Product render layout of heights	(a) keep the render-only layout with heights drawn closer together; (b) render at installed heights	(a), with the note on the hero render	Renders only, not the build	docs/REVIEW.md , 2026-09-26, item 1
6	Status light	(a) one low-duty LED behind a lens in the shield front, with an open-bottom slot so the shield still comes off; (b) no light	(a)	Controller wiring and the shield, if adopted	docs/REVIEW.md , 2026-09-26, item 2
7	Clear enclosure lid	(a) accept a clear-lid box and say so in BOM line 3; (b) grey lid	(a)	Enclosure purchase	docs/REVIEW.md , 2026-09-26, item 3
8	First co-design partner and region	Partner type and region of the first site	None yet; partners are picked per area later	Site survey and installation	WWT-DDR-001, O1
9	Alert language and recipients	Wording beyond the R13 rule, recipients per site	None yet; from co-design	Firmware settings, not the hardware	WWT-DDR-001, O2
10	Data ownership	Who owns the data and who may see it	None yet; from co-design	Upload endpoint, not the hardware	WWT-DDR-001, O3

The 2026-09-26 proposal to add a panel gland on the enclosure top (docs/REVIEW.md , item 4) is overtaken by WWT-DDR-003, P4, which puts every entry in the bottom face, the option that review offered.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The enclosure model, its boss spacing (assumed 166 x 185 mm) and its four-lug kit	They set the internal plate holes and the lug holes in the back plate	WWT-DDR-003, P2 and P3
2	The U-bolt and V-saddle kits fit 48.3 mm pipe, and the depth of the saddle's V	The V sets how far the pole sits behind the plates (40 mm assumed)	WWT-DDR-003, P1
3	The valve has two M4 mounting holes in its back face, or a maker's bracket	Sets the valve holes in the cell plate	WWT-DDR-003, P10
4	The pole-top mount fits 48.3 mm pipe and the panel frame's holes	The panel position and tilt depend on it	WWT-DDR-003, P11
5	The M25 gland grips the 16 mm chlorine sensor body and the M12 gland the 6 mm temperature probe	Gland clamping ranges differ between makers	WWT-DDR-003, P7
6	The push-fit elbows and the inlet fitting are 1/8 BSPT; the hose tail is 3/8 BSP for 12 mm hose	Sets the taps for the cell body	WWT-DDR-003, P9 and P10
7	The light-to-frequency detectors and the 860 nm LED fit a 5 mm bore	Sets the holder bore	WWT-DDR-003, P6
8	The charger's charging window is 0 to 45 °C in its datasheet	Some chargers of this class fix a different window	WWT-DDR-001, D9

■ Value engineering

Value-engineering target: USD 300 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 319 per base unit (USD 19 over the target), from bom/bom.csv (WWT-CAL-001, K1). A pH probe variant site costs about USD 379 to 419. Main cost drivers and savings worth trying:

- The largest lines are the pole with its U-bolts and saddles (USD 35), the controller carrier board (USD 32), the cellular modem (USD 30), the chlorine sensor (USD 25), the enclosure (USD 22), the battery (USD 20) and the sample line fittings (USD 20).
- Making the design constructable added USD 36: the two back plates (USD 10), the lugs and internal plate (USD 8), fixings (USD 6), three more glands and a blanking plug (USD 4), the cell's gasket, studs and lid glands (USD 4), push-fit elbows (USD 2), the shield fixing (USD 1) and the drain hose tail (USD 1).
- Savings worth trying: the LoRaWAN modem variant (about USD 12 instead of USD 30) where a gateway is in range, which alone brings the unit to about USD 301; a box with moulded mounting flanges instead of a separate lug kit (about USD 3); cutting both back plates from one offcut and buying fixings in bulk for several units (about USD 3 to 5); a locally sourced pole or an existing post at sites that have one (up to about USD 25).

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D10: membrane-free graphite chlorine sensor with monthly DPD check, site pH with a pH probe variant, cellular with LoRaWAN as a variant, separate pole, hourly 90 s flush, 0.2 mg/L and 5 NTU alerts, drain to the basin, tapstands first, LiFePO4 with 0 to 45 °C charging, no SwapCell	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	WWT-DDR-001
2026-09-25	D11 to D14: R1 relaxed to ± 0.2 mg/L or ± 25 %; pH probe at sites above pH 7.5 or moving more than 0.2; confirming reading 15 min after a first crossing; footing cast on the survey visit	Amish: "i accept all your recommendations, go with them across all repos."	WWT-DDR-002
2026-09-30	Make the design physically buildable while drawing the build plan; keep open decisions out of the build plan and in this register	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	WWT-DDR-003 (changes made under this instruction, open for review: open decision 1)
2026-10-01	<code>budget_usd</code> is a value-engineering target, not a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens."	<code>.kit/STANDARDS.md</code> section 18; this register